

EXT1 regulates chondrocyte proliferation and differentiation during endochondral bone development

Matthew J. Hilton^{a,b,c,*}, Laura Gutiérrez^{b,c}, Daniel A. Martinez^{b,d}, Dan E. Wells^{b,c}

^aDepartment of Internal Medicine, Division of Bone and Mineral Diseases, Washington University School of Medicine, St. Louis, MO 63110, USA

^bDepartment of Biology and Biochemistry, University of Houston, Houston, TX 77204-5001, USA

^cInstitute for Molecular Biology, University of Houston, Houston, TX 77204-5001, USA

^dConnective Tissue Physiology Laboratory, University of Houston, Houston, TX 77204-5001, USA

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Abstract

Multiple Hereditary Exostoses (MHE) is an autosomal dominant skeletal disorder most frequently caused by mutations in the *EXT1* gene. MHE affects proper development of endochondral bones, such that all affected individuals present with exostoses adjacent to the growth plate of long bones, while some individuals exhibit additional bone deformities. EXT1 functions as a heparan sulfate (HS) co-polymerase, and when defective causes improper elongation of glycosaminoglycan side chains on core proteins of HS proteoglycans. Although analysis of heterozygous *EXT1*-deficient mice has failed to reveal any significant gross morphological variations in skeletal development, significant alterations in molecular signaling occur in the developing long bones. Our results indicate that defects in EXT1 and the resulting reduction in HS lead to enhanced Indian Hedgehog diffusion causing an increase in chondrocyte proliferation and delayed hypertrophic differentiation. © 2004 Elsevier Inc. All rights reserved.

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Introduction

Multiple Hereditary Exostoses (MHE) is an autosomal dominant skeletal disorder that is caused by mutations at one of at least two distinct loci: 8q24.1 (*EXT1*) [1] and 11p11–12 (*EXT2*) [2]. Most cases of MHE, about 70%, are caused by mutations in the *EXT1* gene. Individuals with MHE present with exostoses, or cartilaginous benign bone tumors, located at the growth plates of endochondral bones, and additionally can display bone deformities, joint fusions, and short stature (OMIM 133700). EXT1 is a member of the EXT-related gene family that is composed of *EXT1* [3], *EXT2* [4], *EXTL1* [5], *EXTL2* [6,7], and *EXTL3* [7,8]. The *EXT1* gene encodes a type II transmembrane protein with

glycosyltransferase activity, and specifically functions as a heparan sulfate (HS) co-polymerase with EXT2 in a hetero-oligomeric complex [9,10]. The EXT1/EXT2 complex elongates glycosaminoglycan (GAG) side chains on core proteins of heparan sulfate proteoglycans (HSPGs).

Cell surface HS and HSPGs are known to play critical roles in the extracellular matrix as co-receptors for signaling molecules, modulators of protein transport, stabilizers for cell–cell adhesion, and participate in many other aspects of cellular communication [11]. For example, HS and HSPGs are known to be important modulators of FGF/FGF receptor interaction and specificity [12,13], the organization of Wnt proteins at the cell surface [14], the interactions between BMP dimers and their receptors [15,16], the interaction between Noggin and the cell surface [17], and in maintaining the proper morphogen gradient and diffusion of Hedgehog (Hh) during development [18].

Since a haploinsufficiency of EXT1, and the correlated reduction of HS, are implicated in the bone disorder of MHE, impairments in one or more of the aforementioned

* Corresponding author. Barnes-Jewish Hospital, North, Yalem Research Room 715, Campus Box 8301, 216 S. Kingshighway, St. Louis, MO 63110, USA. Fax: +1 314 454 5047.

E-mail address: mhilton@im.wustl.edu (M.J. Hilton).

signaling pathways could be responsible for some or all of the phenotypes associated with MHE. The process of endochondral ossification is tightly regulated by many of these signaling pathways. Indian hedgehog (Ihh) is a master regulator of chondrocyte proliferation and differentiation during endochondral bone development. It is expressed early near the center of skeletal elements, and later becomes restricted to prehypertrophic and hypertrophic chondrocytes [19–21]. Ihh signals regulate the onset of hypertrophic differentiation via an indirect regulation of parathyroid hormone-related peptide, PTHrP, expressed in the periarticular region. High concentrations of PTHrP maintain immature chondrocytes in a proliferative state. When PTHrP concentrations fall below a threshold level, chondrocytes initiate hypertrophic differentiation and begin to express *Ihh*. This indirect regulation controls the distance from the periarticular region at which hypertrophic differentiation occurs [21]. Ihh has also been shown to positively regulate chondrocyte proliferation independent of its role in controlling differentiation by acting directly through proliferating chondrocytes [22].

In order to understand the molecular details of MHE, Lin et al. [23] disrupted the primary locus for MHE, *EXT1*, through homologous recombination in mouse embryonic stem (ES) cells. Since the *EXT1* null (*EXT1* $-/-$) mice are embryonic lethal at E8.5, skeletal analysis could not be performed. However, initial observations by Lin et al. [23] demonstrated that heterozygous *EXT1* mice (*EXT1* $+/-$) live to adulthood and display no gross skeletal phenotype during post-natal development. Subsequent analyses of these adult heterozygous mice have demonstrated mild reductions in humerus and femur bone density as compared to wild-type bones (unpublished data). In addition, cytological examination of the growth plates of adult *EXT1* $+/-$ long bones show an increase in chondrocyte numbers within the columnar growth plate, although the overall length of long bones are comparable to those of wild-type mice (unpublished data). Given the mild skeletal phenotype in the adult *EXT1* $+/-$ mice, we examined embryonic bone development for significant phenotypic changes, as well as changes in the molecular mechanisms regulating endochondral ossification. To examine the role *EXT1* plays during endochondral bone development, we have analyzed bone length and morphology, chondrocyte proliferation, the expression of specific markers of chondrocyte differentiation, as well as the involvement and interaction of signaling pathways, specifically the Ihh/PTHrP negative feedback loop.

Materials and methods

Animal husbandry

EXT1 mouse strains were previously described [23]. Mice were used in accordance to guidelines established by the animal care facilities at the University of Houston.

Skeletal preparations

Skeletal preparations were performed on E14.5 and E18.5 mouse embryos according to a modified protocol based on McLeod [24]. Briefly, embryos were fixed in 95% ethanol, treated with acetone, stained with an alizarin red/alcan blue mixture (1 volume 0.3% Alcian Blue 8GS; 1 volume 0.1% Alizarin Red S type; 1 volume glacial acetic acid; 17 volumes 70% ethanol), cleared in 0.5% or 1% potassium hydroxide, and transferred through 20%, 50%, and 80% glycerol solutions in 0.5% or 1% potassium hydroxide.

In situ hybridization

Paraffin-embedded tissue sections (6 μ m) were prepared from wild type and *EXT1* $+/-$ embryonic day 16.5 (E16.5) mouse limbs after overnight fixation in 4% paraformaldehyde at 4°C. Antisense digoxigenin-labeled riboprobes were generated using the MaxiScript In Vitro Transcription kit (Ambion) with a digoxigenin RNA labeling mix (Roche). Plasmid constructs used for probe preparation were as follows: *collagen 2a1* [25], *collagen 10a1* [26], *indian hedgehog* [19], and *patched* [27]. Prior to hybridization, sections were deparaffinized in xylene, taken through an ethanol rehydration series, postfixed in 4% paraformaldehyde, rinsed in PBS, and treated with 10 μ g/ml Proteinase K and 0.1 M triethanolamine with acetic anhydride for 10 min each followed by PBS rinses. Slides were prehybridized at 65°C [50% formamide, 5 $\times$ SSC, 2% SDS, 2% Boehringer Blocking Reagent (BBR), 250 μ g/ml tRNA, and 100 μ g/ml heparin] for 3 h and hybridization was then carried out at 65°C for at least 14 h. Following washes and a 1-h blocking step in 20% heat-inactivated sheep serum and 2% BBR, slides were treated with 1:2500 dilution of anti-digoxigenin-AP fab fragments (Roche) overnight at 4°C. Slides were washed and finally incubated in a NBT/BCIP solution (125 μ g/ml BCIP and 250 μ g/ml NBT in NTM) for 3–5 h at room temperature in order to detect a color reaction.

Immunohistochemistry and BrdU analysis

Slides containing comparable sections from E16.5 mouse limbs were used in the immunohistochemistry procedures according to the Zymed HistoMouse-SP Kit (AEC) and Zymed BrdU staining protocols with minor modifications. Prior to peroxidase quenching, slides used for antibody staining of Ihh were incubated with 1 mg/ml hyaluronidase at 37°C for 15 min. On the day of harvesting E16.5 embryos for BrdU analysis, the mother was given an intraperitoneal injection of 0.1 ml per 10 g of body weight of a 10 mg/ml bromodeoxyuridine (BrdU) (Sigma): 1.2 mg/ml fluorodeoxyuridine (FdU) (Sigma) stock solution. At least three different sections were counted for four wild type and four *EXT1* $+/-$ E16.5 mouse tibias in order to calculate BrdU indices.

Quantitative RT-PCR

For a specific gene of interest, 5'-Taqman hydrolysis Q-PCR assays were performed using an MX-4000 real-time Q-PCR machine (Stratagene). For each Q-PCR assay, the sequences for forward and reverse oligonucleotide primers (20–24 nt), fluorescent reporter probes (20–25 nt) containing a 5'-FAM, -HEX, or -Cy5 fluorescent reporter, and 3'-BHQ 1–3 quenchers™ (BioSource Int.) have been generated using Beacon Designer 2 Taqman multiplexing software (Premier Biosoft). Standards were generated using synthetic-DNA (sDNA) amplicon oligonucleotides (BioSource Int.) spanning the entire Q-PCR amplification (75–150 nucleotides) region. A standard curve methodology was performed to quantitate the specific gene targets of interest. Specific standard aliquots (10^8 to 10^1 copies of cDNA) were used to generate a linear curve ($>R^2 = 0.995$) with a Q-PCR efficiency (slope) of -3.2 to -3.3 to determine the cDNA copy number in three replicates. Each Q-PCR reaction consists of 12 min at 95°C to activate the hot-start Taq polymerase (HotStar Master Mix, Qiagen) followed by 40 cycles of two-step thermocycling (45 s at 94°C and 12 s at 60°C).

Results

Skeletal analysis of *EXT1* heterozygous mutant mouse embryos

Wild-type and *EXT1*^{+/-} mouse embryos were cleared and stained with alcian blue and alizarin red in order to analyze skeletal morphology, bone length, and percent ossification of endochondral bones during two stages of embryonic skeletal development, E18.5 and E14.5. E18.5 embryos were analyzed primarily for phenotypes that would be consistent with MHE. *EXT1*^{+/-} skeletons did not display any observable bone deformities, occurrence of exostoses, or changes in axial or appendicular skeletal structure. Overall, the *EXT1*^{+/-} skeletons do not display a significant change in skeletal morphology as compared to wild-type embryos (data not shown). In addition, E14.5 embryos were similarly analyzed in order to detect potential changes in proliferation during early stages of chondrogenesis, prior to ossification and calcification of the endochondral bones. Again, the *EXT1*^{+/-} embryos showed no significant differences in their skeletal morphology at this early stage of chondrogenesis when compared to wild-type mice (data not shown).

To further analyze the size and degree of ossification of long bones for *EXT1*^{+/-} and wild-type mice, E18.5 humeri, femurs, and tibiae were dissected from mice used for skeletal preparations. These bones were catalogued using digital imaging equipment and measured with Image-Pro Express software. *EXT1*^{+/-} long bones display no significant change in length or ossification as

compared to wild-type bones, although *EXT1*^{+/-} bones show greater variation in both length and percent ossification (Table 1).

Proliferative and hypertrophic marker analysis in *EXT1* heterozygous mutant long bones

In order to investigate potential changes in proliferation or hypertrophic differentiation, in situ hybridization was performed using riboprobes for *collagen 2a1*, a marker for proliferating chondrocytes, and *collagen 10a1*, a marker for hypertrophic chondrocytes. At E16.5, *EXT1*^{+/-} proximal tibia sections exhibit a moderate increase, $6.6 \pm 1.1\%$ ($n = 8$; $P < 0.05$) in the expression domain of *collagen 2a1* as compared to wild-type littermates. The *collagen 2a1* expression domain was measured from the periarticular region to the final *collagen 2a1* expressing cells near the hypertrophic region (Figs. 1A,B). *EXT1*^{+/-} embryos also display a reduction in the *collagen 10a1* expression domain in proximal tibia growth regions (Figs. 1C,D). At least three sections were analyzed from eight animals of each genotype ($n = 8$). Similar results were also obtained from forelimb growth plates of E16.5 *EXT1*^{+/-} and wild-type animals (data not shown). In addition, quantitative RT-PCR results demonstrate a near $44.2 \pm 3.0\%$ ($n = 7$; $P < 0.01$) reduction in *collagen 10a1* expression in E18.5 whole tibiae. As a positive control, *EXT1* expression was quantified in wild-type and *EXT1*^{+/-} tibiae. *EXT1*^{+/-} tibiae show over a $35.1 \pm 5.6\%$ ($n = 7$; $P < 0.01$) reduction in *EXT1* expression as compared to wild-type tibiae (Fig. 2). Similar results were also obtained when analyzing E18.5 humeri from both *EXT1* $\pm$ and wild-type mice (data not shown).

An increase in the proliferative region, as seen in the expanded *collagen 2a1* expression domain, suggests that *EXT1*^{+/-} embryos display an increase in chondrocyte proliferation and/or delay in hypertrophic differentiation. The reduction in the *collagen 10a1* expression domain, which is also located at a greater distance from the periarticular region, provides further support for an apparent change in the maturation of *EXT1*^{+/-} chondrocytes.

Table 1
Skeletal analysis of E18.5 $+/+$ and $+/-$ mice

	Bone length (mm)	% Ossification
Tibia $+/+$	4.35 ± 0.10	49.58 ± 2.54
Tibia $+/-$	4.24 ± 0.33	49.60 ± 5.54
Femur $+/+$	4.00 ± 0.10	42.91 ± 1.38
Femur $+/-$	4.04 ± 0.27	44.74 ± 4.84
Humerus $+/+$	4.42 ± 0.15	50.28 ± 2.16
Humerus $+/-$	4.49 ± 0.31	51.05 ± 4.25

Statistical analysis revealed that *EXT1*^{+/-} tibiae, femurs, and humeri do not present a significant change in their lengths or % ossification as compared to wild-type ($+/+$) bones ($n = 12$).

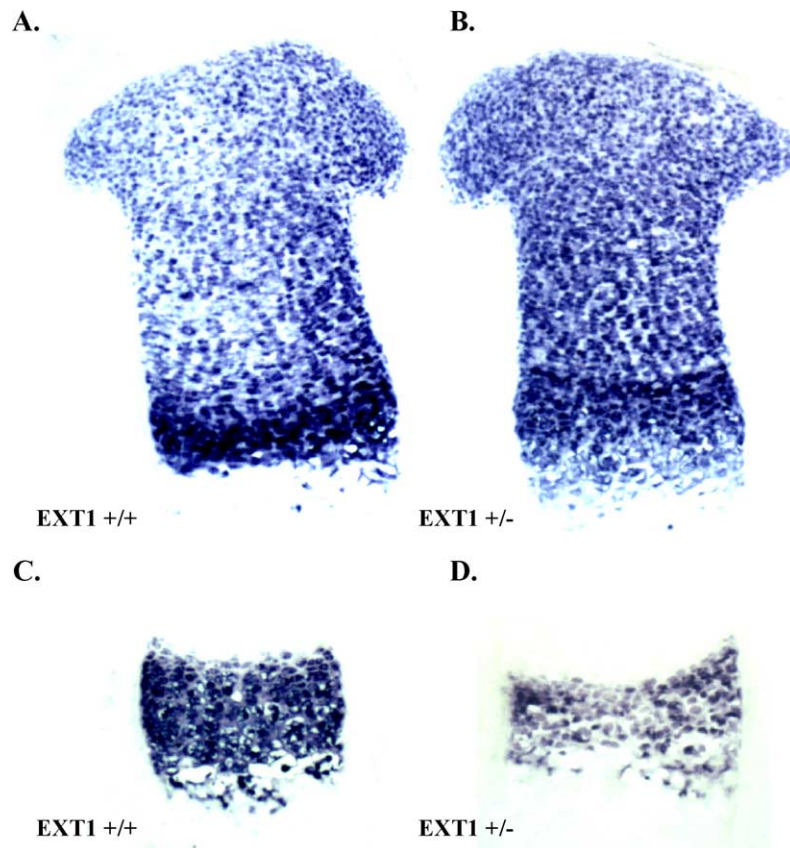


Fig. 1. Proliferating and hypertrophic chondrocyte marker expression analysis. Antisense digoxigenin-labeled RNAs for either *collagen 2a1* (A, B) or *collagen 10a1* (C, D) were in situ hybridized to E16.5 proximal tibia sections of either *EXT1*^{+/+} (A, C) or *EXT1*^{+/-} (B, D) mice. The expression domain of *collagen 2a1*, a proliferative chondrocyte marker, is expanded in *EXT1*^{+/-} (B) as compared to *EXT1*^{+/+} (A) tibias. Conversely, the expression domain of *collagen 10a1*, a hypertrophic chondrocyte marker, is restricted in *EXT1*^{+/-} (D) as compared to *EXT1*^{+/+} (C) tibias.

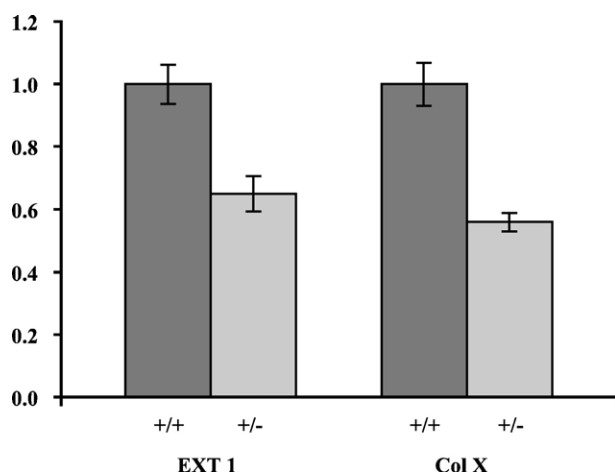


Fig. 2. Quantitative RT-PCR analysis of *EXT1* and *collagen 10a1* expression in whole tibias. *EXT1*^{+/-} (HET) mice show a $35.1 \pm 5.6\%$ reduction in *EXT1* expression in whole tibias of E18.5 embryos ($n = 7$; $P < 0.01$). Additionally, *collagen 10a1* (Col X) expression is reduced by $44.2 \pm 3.0\%$ as compared to wild-type (WT) tibias ($n = 7$; $P < 0.01$). Wild-type gene expression is normalized to 1, and *EXT1*^{+/-} gene expression is shown as changes in expression relative to wild type.

EXT1 affects chondrocyte proliferation and differentiation during endochondral bone development

The enlarged proliferative zone (*collagen 2a1* expression) and reduced expression of *collagen 10a1* suggested that *EXT1*, either directly or indirectly, affects chondrocyte proliferation and/or differentiation. To examine the effect of *EXT1* on chondrocyte proliferation, bromodeoxyuridine (BrdU) incorporation was monitored in wild-type and *EXT1*^{+/-} E16.5 proximal tibia sections using immunohistochemistry (Fig. 3). Calculations were made by dividing the number of BrdU-positive chondrocytes by the total number of proliferating chondrocytes, those cells extending from the articular region to cells preceding hypertrophy. BrdU labeling in *EXT1*^{+/-} embryos increases $10.9 \pm 1.9\%$ ($n = 4$; $P < 0.01$) as compared to wild-type littermates (Fig. 3C). In order to confirm lengthening of the proliferative region and delay in hypertrophic differentiation of *EXT1*^{+/-} endochondral bones, the distance from the periarticular region to the final BrdU positive chondrocytes was measured in wild-type and *EXT1*^{+/-} proximal tibia sections. The distance to the final BrdU-positive proliferating chondrocytes is increased by $7.3 \pm 1.7\%$ ($n = 6$; $P < 0.01$) in *EXT1*^{+/-} embryos as compared to wild-type

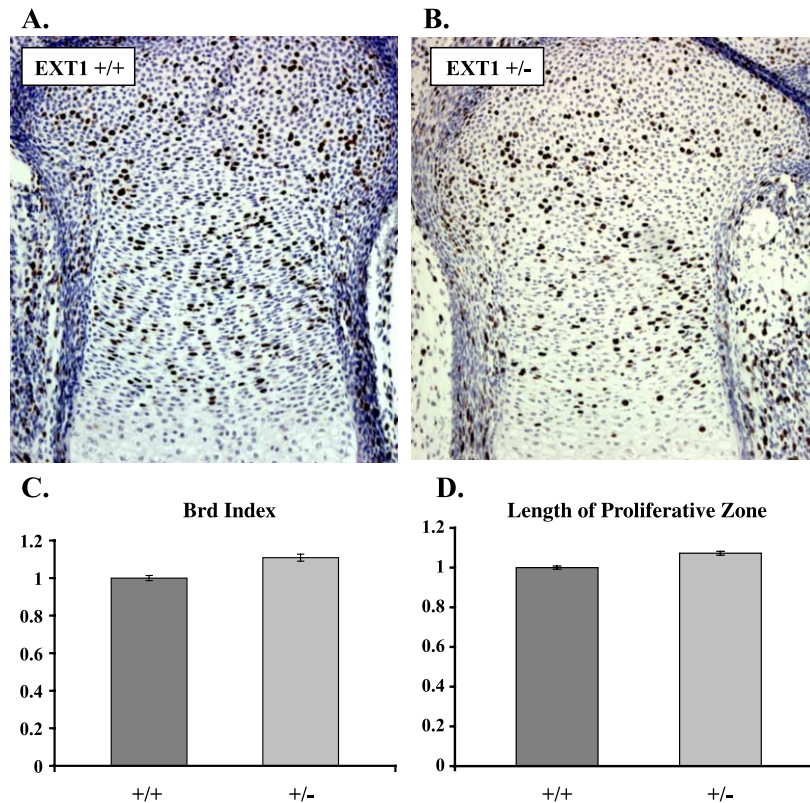


Fig. 3. Analysis of chondrocyte proliferation and differentiation using BrdU labeling. BrdU incorporation into proliferating chondrocytes was monitored for E16.5 proximal tibia sections of either *EXT1*^{+/+} (A) or *EXT1*^{+/-} (B) mice. *EXT1*^{+/-} (HET) E16.5 proximal tibia sections show a $10.9 \pm 1.9\%$ ($n = 4$; $P < 0.01$) increase in the percentage of BrdU-positive nuclei as compared to wild-type (WT) sections (C). *EXT1*^{+/-} (HET) mice display a $7.3 \pm 1.7\%$ ($n = 6$; $P < 0.01$) increase in the distance to the final BrdU-positive chondrocyte (D). The proportion of BrdU-positive nuclei and wild-type measurements were normalized to 1. Sections are counterstained with hematoxylin (blue).

littermates (Fig. 3D). This increase in the proliferative zone as measured by BrdU analysis confirms the data obtained from *collagen 2a1* expression experiments. Collectively, these data provide evidence for a role of EXT1 in regulating both chondrocyte proliferation and initiation of hypertrophic differentiation.

EXT1 heterozygotes display altered Indian Hedgehog signaling during endochondral bone development

To determine the molecular mechanisms responsible for increased proliferation and delayed hypertrophic differentiation of chondrocytes in *EXT1*^{+/-} embryos, we first examined potential defects in *Ihh* signaling. Given the well-documented role *Ihh* plays in chondrocyte proliferation and differentiation and the effect *ttv*, the *Drosophila* *EXT1* homologue, has on Hh diffusion and signaling [18], we performed immunohistochemical and in situ hybridization analyses in order to determine the exact effect EXT1 has on *Ihh* signaling during endochondral bone development.

First, wild-type and *EXT1*^{+/-} E16.5 proximal tibia sections were incubated with the 5E1 monoclonal antibody against the N-terminal peptide of *Ihh*. Comparison of wild-type and *EXT1*^{+/-} sections generally show a

more diffuse staining pattern in *EXT1*^{+/-} sections for *Ihh*. (Figs. 4A,B). Despite the increased domain of *Ihh* staining in *EXT1*^{+/-} sections, in situ hybridization data revealed that *Ihh* expression is restricted to a more discrete domain within the prehypertrophic region (Figs. 4C,D). This reduction in the *Ihh* expression domain of *EXT1*^{+/-} sections makes the increased *Ihh* diffusion more significant.

Increased *Ihh* activity has been shown to increase chondrocyte proliferation and delay the initiation of hypertrophic differentiation [20–22]. To further address the effect of EXT1 on *Ihh* signaling, in situ hybridization was performed for the *Ihh* transcriptional target gene, *patched*. Consistent with the observations of increased *Ihh* diffusion in *EXT1*^{+/-} embryos, *patched* expression is enhanced in the perichondrium flanking the prehypertrophic region, and also expanded to proliferating chondrocytes very near the periarticular region of E16.5 proximal tibia sections (Fig. 5). The increase in *Ihh* diffusion, enhanced and expanded *patched* expression suggests a molecular mechanism to explain the increased proliferation and delayed hypertrophic differentiation seen in *EXT1*^{+/-} endochondral bones during embryonic development. However, alterations in other signaling pathways such as the FGFs, BMPs, or Wnts could

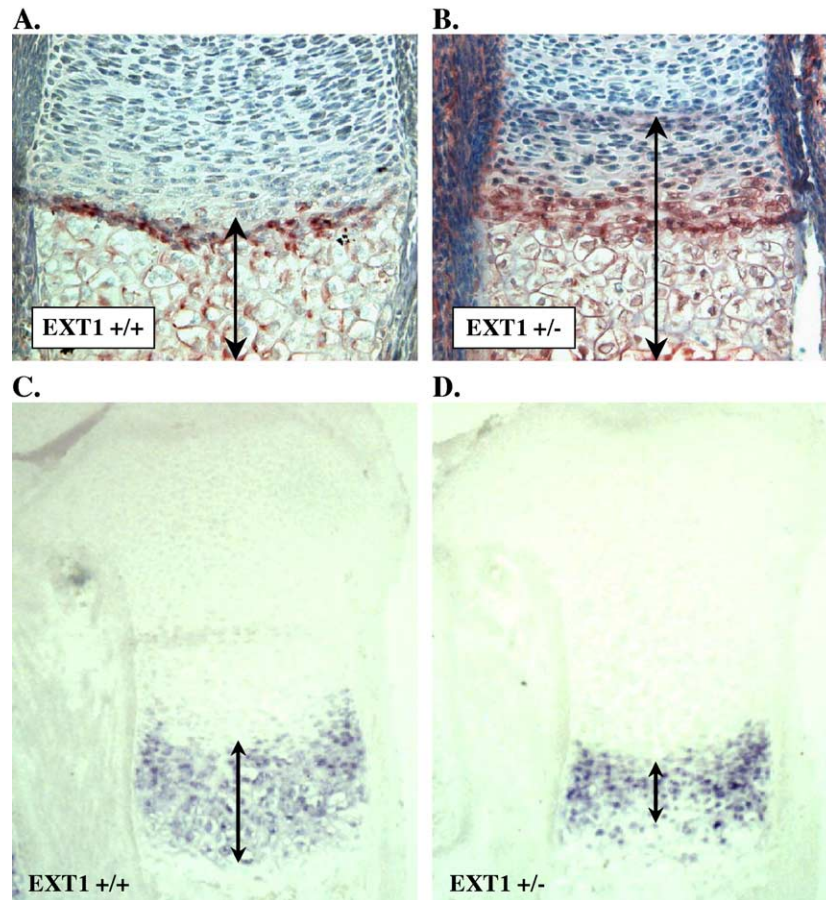


Fig. 4. Analysis of Ihh diffusion and *Ihh* expression in proximal tibia sections. Immunohistochemical localization of Ihh protein using 5E1 antibody to E16.5 proximal tibia sections of either *EXT1*+/+ (A) or *EXT1*+/- (B) mice. *EXT1*+/- (B) sections display an expanded Ihh protein domain (Red) as compared to wild-type (A) sections. Antisense digoxigenin labeled RNAs for *Ihh* were in situ hybridized to E16.5 proximal tibia sections of either *EXT1*+/+ (C) or *EXT1*+/- (D) mice. The *Ihh* expression domain is reduced in *EXT1*+/- (D) tibia sections as compared to wild-type (C) sections. Arrows indicate the extent of primary Ihh immunohistochemical staining (A, B) or the extent of *Ihh* expression domains (C, D) in comparable wild-type and *EXT1*+/- sections.

additionally cause altered chondrocyte proliferation and maturation.

Discussion

Mutations in both *EXT1* or *EXT2* have been shown to cause MHE. Since both *EXT1* and *EXT2* function as HS copolymerases, it is believed that defects in HS biosynthesis cause the characteristic skeletal phenotypes associated with the disease. The process of endochondral bone development is partially controlled by signaling molecules that utilize HS and HSPGs to facilitate ligand–receptor interactions, control proper protein diffusion, and maintain association of proteins at the cell surface or within the extracellular matrix. Since *EXT1* and *EXT2* are expressed in proliferating chondrocytes, prehypertrophic chondrocytes, and osteoblasts [28], potential defects in signaling might occur at almost any stage of endochondral ossification. To address the effect *EXT1* has on endochondral bone development, we analyzed *EXT1*+/- mutant mice for defects in skeletal morphology, chondrocyte proliferation and differentiation, and alterations in gene

expression and protein distribution for molecules of the Ihh signaling pathway during embryonic development.

EXT1 regulates *Ihh* diffusion and subsequently alters chondrocyte proliferation and differentiation

Results from *EXT1*+/- limbs indicate an expansion of Ihh distribution (Figs. 4A,B), while the *Ihh* expression domain is restricted primarily to the prehypertrophic chondrocytes rather than to both prehypertrophic and hypertrophic chondrocytes as seen in wild-type embryos. Additionally, Ihh staining seems to be more abundant in regions of the perichondrium, flanking the expanded Ihh domain within the hypertrophic and proliferating chondrocytes (Figs. 4C,D). Similar results were also observed in E16.5 *EXT1*+/- and wild-type forelimbs (data not shown). Increased Ihh diffusion and activity was confirmed by analyzing the Ihh target gene, *patched*. *EXT1*+/- embryos show enhanced *patched* expression in the perichondrium flanking the Ihh expression domain, and also an expanded domain of *patched* expression in immature chondrocytes extending from the prehypertrophic chondrocytes to a

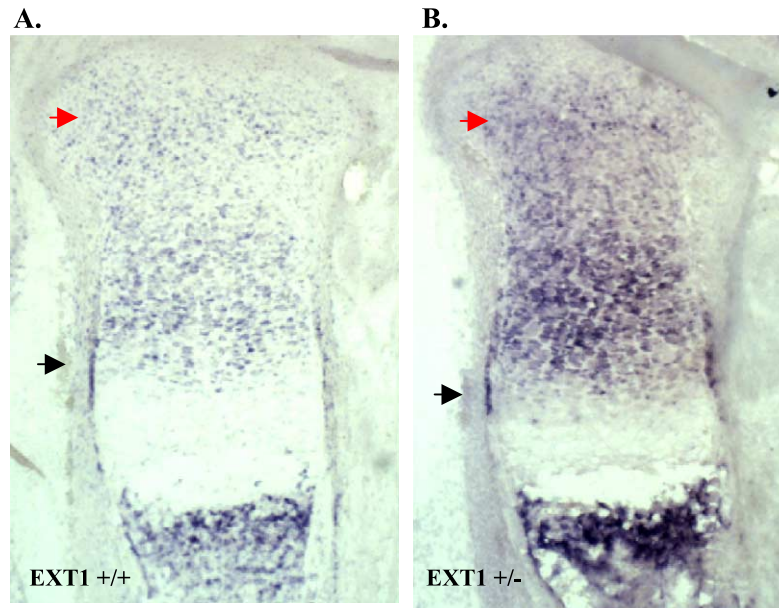


Fig. 5. Analysis of *patched* expression in endochondral bones. Antisense digoxigenin-labeled RNAs for *patched* were in situ hybridized to E16.5 proximal tibia sections of either *EXT1*^{+/+} (A) or *EXT1*^{+/-} (B) mice. *EXT1*^{+/-} (B) proximal tibia sections show enhanced *patched* expression in the perichondrium (black arrow heads) and expanded expression in the proliferating chondrocytes from the prehypertrophic zone to the periarticular region (red arrow heads) as compared to wild-type (A) sections.

region very near the periarticular perichondrium (Fig. 5). Similar results in delayed differentiation and increased proliferation are seen when *Ihh* is misexpressed within endochondral bones using viral infection or genetic manipulation [20,22]. Additionally, the expanded *collagen 2a1* expression, restricted and reduced *collagen 10a1* expression, expanded *patched* expression, and increased BrdU indices found in *EXT1*^{+/-} embryos are consistent with various experiments describing the affects of ectopic *Ihh* signaling on endochondral bone development [20–22,29,30]. We believe that reduced HS biosynthesis increases *Ihh* diffusion and signaling throughout prehypertrophic/hypertrophic and many immature chondrocytes causing increased proliferation and delayed differentiation seen in *EXT1*^{+/-} embryonic long bones.

Studies using the *Drosophila* *EXT1* homologue, *ttv*, mutant suggested that Hh diffusion and signaling were diminished due to complete disruption of HS biosynthesis [18,31]. Additionally, Lin et al. [23] demonstrated an inability of *Ihh* to associate with the cell surface of the visceral endoderm in *EXT1*^{-/-} embryos. These data are in contrast to our results showing an increase in *Ihh* diffusion and signaling during endochondral ossification due to the haploinsufficiency of *EXT1*. The most likely explanation for this discrepancy is the clear differences in the overall levels of HS production. Our data and recent data generated by Andrea Vortkamp [32] has lead us to suggest that *Ihh* requires HS for association at the cell surface, as well as for transportation across multiple cell diameters, but might also act to limit *Ihh* diffusion in normal concentrations. Therefore, *EXT1*^{+/-} mice show regions of expanded *Ihh* diffusion and activity due to a near 50% reduction in HS,

while *ttv* mutants and early *EXT1*^{-/-} embryos (no HS production) show restricted Hh and *Ihh* diffusion and a complete lack of association at the cell surface. Another mutant mouse carrying hypomorphic alleles for *EXT1* produces only 18% of the normal levels of HS. Analyses of these mice have demonstrated even greater range of *Ihh* diffusion and signaling as compared to *EXT1*^{+/-} mice [32]. Additionally, many of the findings we have reported here describing the effect *EXT1* has on *Ihh* signaling is supported by recent data generated from the aforementioned *EXT1* hypomorph [32].

Molecular consequences of EXT1 haploinsufficiency and MHE pathology

Signals between the perichondrium and chondrocytes from areas flanking the hypertrophic and periarticular regions provide positional cues establishing the vertical columns of chondrocytes, which ensure proper longitudinal bone growth. Disruptions in communication between the perichondrium/periosteum, or bone collar, and chondrocytes can drastically alter endochondral bone development. Stickens et al. [28] proposed a model to explain the molecular mechanisms underlying exostosis formation. In this model, individuals haploinsufficient for either *EXT1* or *EXT2* would obtain a second somatic mutation in either an *EXT* gene, or perhaps a third related gene. The subsequent mutant clone would proliferate giving rise to many mutant borderline chondrocytes (near the perichondrium) or mutant perichondral cells. These mutant cells would disrupt signaling between the perichondrium and chondrocytes due to the complete loss of HS biosynthesis. The localized mutations

could result in a temporal and spatial disruption in the differentiation of the bone collar, allowing chondrocytes to escape from negative growth regulation imposed by the perichondrium. Alternatively, our results indicate that the enhancement in *Ihh* stimulation of chondrocytes and perichondral cells might lead to increased proliferation and a disruption in the differentiation process without the necessity of a secondary mutation. These alterations by themselves, or combined with potential secondary somatic mutations, likely contribute to the MHE pathology.

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